Lakmali Nadeesha Kumari

Postdoctoral Researcher – University of California, Davis

lrathukohe@ucdavis.edu | GitHub | LinkedIn | Phone: +1 346 702 8186

Research Interests

Agentic AI and data-efficient machine learning for biomedical data: multi-agent vision–language systems for digital neuropathology, foundation-model adaptation and difficulty-aware segmentation of gigapixel histopathology, active and self-supervised learning under scarce annotation, and deep generative models for discovering disease-progression subtypes in Alzheimer's disease.

Academic Appointments

Postdoctoral Researcher Oct. 2026 – Present
UC Davis, Depts. of Electrical & Computer Engineering and Pathology & Laboratory Medicine *Davis, CA*
Mentors: Prof. Chen-Nee Chuah and Prof. Brittany N. Dugger

- Developing agentic vision–language AI and foundation-model methods for scalable, quantitative assessment of Alzheimer's disease and related dementia (AD/ADRD) neuropathology in whole-slide images.
- Building generalizable digital-neuropathology workflows across institutions, scanners, and stains; contributing to NIH grant proposals in AI for neuropathology.

Education

Ph.D., Electrical and Computer Engineering (GPA 3.8/4.0) Aug. 2021 – Sep. 2026
University of Kentucky Advisor: Prof. Sen-ching Samson Cheung *Lexington, KY*
Dissertation: *Resource-Efficient Machine Learning Through the Balance of Prior Knowledge and Learned Structure*

B.Sc. Eng., Electrical and Electronic Engineering 2013 – 2017
University of Peradeniya (Washington Accord accredited) *Peradeniya, Sri Lanka*

Research Experience

Graduate Research Assistant, Multimedia Information Analysis (MIA) Lab Aug. 2022 – Sep. 2026
University of Kentucky Advisor: Prof. S.-C. S. Cheung *Lexington, KY*

- **Difficulty-aware histopathology segmentation:** Dynamic Focal Attention, a learned per-class attention bias improving difficult classes by up to 15.2 Dice points with no separate pre-training stage; validated on BDSA brain WSIs, BCSS, and CRAG [C2].
- **Foundation models for brain pathology:** adapted SAM-Path for multicompartment neuroanatomic segmentation of autopsy brain sections from the Brain Digital Slide Archive [A1].
- **Alzheimer's progression subtypes:** CONTINUUM, amortized variational inference over continuous subtype trajectories; five reproducible subtypes across two independent autopsy cohorts with replicated genetic associations (with Dr. P. T. Nelson, UK Neuropathology) [U2].
- **Annotation-efficient segmentation:** task-aligned active learning reaching 98.7% of fully supervised Dice with 38.4% of labeled ISIC-2017 data, resolving the cold-start problem [C1].
- **Behavioral video analysis for autism:** gaze-feature extraction pipeline for the UC Davis MIND Institute video cohort (with Dr. C.-N. Chuah and Dr. S. Ozonoff).

Graduate Research Assistant, Augmented Intelligence for Smart Manufacturing Lab Aug. 2021 – Aug. 2022
University of Kentucky Advisor: Prof. Peng Wang *Lexington, KY*

- Conditional invertible neural network for amortized Bayesian estimation of battery degradation, cutting per-trajectory inference from 690 s (MCMC) to 0.2 s while improving prognosis [J3].

Industry Research Experience

Applied Scientist Intern (2 summers) Jun.–Aug. 2024, May–Aug. 2025
Amazon, Merch on Demand *Seattle, WA*

- 2025: generative-AI product visualization pipeline (diffusion models, LoRA, LLM-as-judge evaluation).
- 2024: multi-modal product classification with CLIP, automated active learning, and LLM pseudo-labeling.

Applications Engineer II, Synopsys, Sri Lanka (EDA static, CDC/RDC and constraint verification) Apr. 2018 – Aug. 2021

Publications

Peer-Reviewed Conference Papers

[C2] **R.M.L.N. Kumari**, S.-C. S. Cheung. "Learning Class Difficulty via Dynamic Focal Attention for Histopathology Segmentation." *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, LNCS vol. 16891, Springer, 2026. [PDF]

[C1] **R.M.L.N. Kumari**, C. T. Bandara, C.-N. Chuah, S.-C. S. Cheung. "A Warmer Start to Active Learning with Adaptive Gaussian Mixture Models for Skin Lesion Segmentation." *IEEE International Conference on Image Processing (ICIP)*, pp. 2247–2252, 2025. [DOI]

Journal Articles

[J3] **R.M.L.N. Kumari**, P. Wang. "Efficient Stochastic Parametric Estimation for Lithium-Ion Battery Performance Degradation Tracking and Prognosis." *Journal of Manufacturing Systems*, vol. 75, pp. 270–277, 2024. [DOI]

[J2] **R.M.L.N. Kumari**, C. T. Bandara, M. B. Dissanayake. "Sylvester Matrix Based Similarity Estimation Method for Automation of Defect Detection in Textile Fabrics." *Journal of Sensors*, vol. 2021, 2021. [DOI]

[J1] C. T. Bandara, **R.M.L.N. Kumari**, D. H. S. Maithripala, D. A. A. C. Ratnaweera. "Vehicle-Fixed-Frame Adaptive Controller and Intrinsic Nonlinear PID Controller for Attitude Stabilization of a Complex-Shaped Underwater Vehicle." *Journal of Mechatronics and Robotics*, vol. 4, pp. 254–264, 2020. [DOI]

Under Review and In Preparation

[U2] **R.M.L.N. Kumari**, J. Zhu, R. K. Shahidehpour, P. T. Nelson, S.-C. S. Cheung. "CONTINUUM: A Variational Deep-Generative Reformulation of Subtype and Stage Inference Reveals Reproducible, Genetically Distinct Neuropathological Trajectories of Alzheimer's Disease Across Independent Autopsy Cohorts." In preparation for *Neurobiology of Disease*.

[U1] **R.M.L.N. Kumari**, C. T. Bandara, S.-C. S. Cheung. "Not All Steps Matter: Temporally Adaptive Spectral Control for Concept Removal in Diffusion Models." In preparation for *IEEE/CVF CVPR 2027*.

Published Abstracts

[A1] T. Pearce, **L. Nadeesha**, A. Neltner, *et al.* "Multicompartment Neuroanatomic Segmentation of Autopsy Brain Tissue Sections Using Deep Learning Methodologies." *Journal of Neuropathology & Experimental Neurology*, 84(6):494, 2025 (101st Annual Meeting, American Association of Neuropathologists).

Posters and Presentations

- "SemiSAM_Med: Class-Imbalanced Active Learning in Medical Image Segmentation via a Semi-Supervised SAM Model." ECE Research Symposium, University of Kentucky, 2024. [Poster]
- "Eyes Speak Louder: Computer Vision and Machine Learning Insights into Autism Spectrum Behaviors." Commonwealth Computational Summit, UK Center for Computational Sciences, 2023. [Poster]
- "SAL-SMIS: Self-Supervised Active Learning for Semi-Supervised Medical Image Segmentation." ECE Research Symposium, University of Kentucky, 2023. [Poster]
- "Modeling and Prediction of Lithium-Ion Battery Degradation with CINN for Parameter Estimation." Engineering Dean's Advisory Council, University of Kentucky, 2023. [Poster]

Teaching and Mentoring

Graduate Teaching Assistant, Electrical & Computer Engineering, University of Kentucky — Jan. 2022 – Dec. 2023
Instructor (Teaching Assistant), Electrical & Electronic Engineering, University of Peradeniya — Nov. 2017 – Apr. 2018
Volunteer Mathematics & Science Teacher, "Arunella" national social project, Sri Lanka — 2014

Service and Leadership

Assistant Director, Graduate Society of Women Engineers (GradSWE), University of Kentucky — 2023 – 2024
Student Volunteer, 12th IEEE International Conference on Industrial and Information Systems (ICIIS) — 2017
Associate Member, Institution of Engineers, Sri Lanka (IESL) — 2016 – Present

Honors and Awards

Synopsys recognition awards for technical contributions — 2019 – 2021
University Colours in Basketball and Hockey, University of Peradeniya — 2014 – 2017
GCE Advanced Level: top 1% of ∼250,000 candidates nationally, Sri Lanka — 2012

Technical Skills

Programming Python, MATLAB, R, C/C++, Bash, SQL; PyTorch, TensorFlow, Hugging Face
Methods Agentic LLM / vision–language pipelines, foundation models (SAM, CLIP), diffusion models, LoRA/PEFT, active & self-supervised learning, variational inference
Biomedical Whole-slide image pipelines, BDSA, TCGA, BCSS, CRAG, ISIC dermoscopy, eye-tracking video
Computing Multi-GPU training, HPC (NSF ACCESS Delta), AWS (SageMaker, Bedrock), Docker, Git, Linux

References

Prof. Sen-ching Samson Cheung
Ph.D. Advisor, Professor of ECE
University of Kentucky
sccheung@ieee.org

Dr. Peter T. Nelson
Director of Neuropathology
University of Kentucky
pnels2@uky.edu

Prof. Daniel Lau
Director of Graduate Studies, ECE
University of Kentucky
dllau@uky.edu